Principal polynomial submodule projections need not almost commute

A counterexample to the question in arXiv:1204.0620

Candidate solution packet — run `fa_banach_001`, lane 2

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Status. Candidate counterexample, likely valid; human review is recommended. In every dimension $n \geq 3$, the projections onto the principal submodules generated by z_1 and $z_1 + z_2$ fail to commute modulo compact operators in each of the Hardy, Bergman, and Drury–Arveson modules on the unit ball.

1 Source question

Douglas and Wang ask whether the projections onto two principal polynomial submodules might always almost commute [1]. Their paragraph begins at the bottom of printed page 7 and continues on printed page 8:

> While it seems inconceivable that $[p] + [q]$ is always closed for polynomials p and q in $\mathbb{C}[z_1, \cdots, z_n]$; here $[\cdot]$ denotes the closure in the Hardy, Bergman or Drury-Arveson modules on the unit ball, it seems quite possible that the projections onto $[p]$ and $[q]$ always almost commute. One thing making the answering of such a question difficult is the fact that $[p] \cap [q]$ is always large containing $[pq]$. One possible way to circumvent
>
> this problem might be to consider the quotient modules $[p]^\perp$ and $[q]^\perp$. We'll have something more to say about them in the next section.

The exact suggestion is:

> For polynomials $p, q \in \mathbb{C}[z_1, \ldots, z_n]$, if $P_{[p]}$ and $P_{[q]}$ are the orthogonal projections onto the closures of the corresponding principal ideals in the Hardy, Bergman, or Drury–Arveson module on the unit ball, is $[P_{[p]}, P_{[q]}]$ always compact?

Here "almost commute" means membership of the commutator in the compact operator ideal $\mathcal{K}$.

2 Setup

Let $\mathcal{H}$ be one of the following Hilbert modules on $\mathbb{B}^n$: the Hardy module $H^2(\partial\mathbb{B}^n)$, the Bergman module $A^2(\mathbb{B}^n)$, or the Drury–Arveson module H^2_n. Each is an orthogonal sum of its finite-dimensional homogeneous pieces,

$$\mathcal{H} = \bigoplus_{m \geq 0} \mathcal{H}_m.$$

The monomials form an orthogonal basis, and for $|\alpha| = m$ their squared norms have the form

$$\|z^\alpha\|^2 = c_m \, \alpha!, \tag{1}$$

where $c_m > 0$ depends on the module and on m, but not on the arrangement of the coordinates of α. In particular, exchanging z_1 and z_2 preserves monomial norms.

For a polynomial r, write $[r]$ for the closure of $r\mathbb{C}[z_1, \ldots, z_n]$ in $\mathcal{H}$, and $P_{[r]}$ for its orthogonal projection.

3 Proof intuition

The source's positive criterion applies when the two boundary zero varieties are disjoint. Two transverse hyperplanes in dimension at least three meet along a nontrivial boundary set, and that common direction lets the same two-dimensional angle recur in every homogeneous degree. Concretely, the factor z_3^{m-1} carries the angle between z_1 and z_1+z_2 unchanged through all degrees. The quotient projections therefore contain infinitely many identical 2×2 blocks, each with commutator norm $1/2$.

4 Counterexample

Theorem 1. *Let $n \geq 3$, and let $\mathcal{H}$ be the Hardy, Bergman, or Drury–Arveson module on $\mathbb{B}^n$. Set*

$$p = z_1, \qquad q = z_1 + z_2.$$

If $P = P_{[p]}$ and $Q = P_{[q]}$, then $[P,Q] \notin \mathcal{K}(\mathcal{H})$. More precisely, $[P,Q]$ has a reducing two-dimensional block of norm $1/2$ in every positive homogeneous degree.

Proof. Put $E = I - P$ and $F = I - Q$, the orthogonal projections onto $[p]^\perp$ and $[q]^\perp$. Since

$$[I-E, I-F] = EF - FE = [E,F],$$

it is enough to prove that $[E,F]$ is not compact.

For every $m \geq 1$, define

$$u_m = z_1 z_3^{m-1}, \qquad v_m = z_2 z_3^{m-1}, \qquad V_m = \operatorname{span}\{u_m, v_m\} \subset \mathcal{H}_m.$$

By (1), u_m and v_m are orthogonal and have equal norm. We use the normalized ordered basis (e_m, f_m) obtained from (u_m, v_m).

The degree-m part of $[z_1]$ is spanned by the degree-m monomials which are divisible by z_1. Thus $u_m \in [z_1]$, while v_m is orthogonal to the whole degree-m part of $[z_1]$. Consequently V_m reduces E and

$$E|_{V_m} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2}$$

Next,

$$u_m + v_m = (z_1+z_2) z_3^{m-1} \in [q].$$

On the other hand, $u_m - v_m$ is orthogonal to $[q] \cap \mathcal{H}_m$. Indeed, if r is a degree-$(m-1)$ monomial other than z_3^{m-1}, then both monomials in $(z_1+z_2)r$ are orthogonal to u_m and v_m; for $r = z_3^{m-1}$, equal norms give

$$\langle u_m - v_m, u_m + v_m \rangle = \|u_m\|^2 - \|v_m\|^2 = 0.$$

By linearity this holds for every homogeneous polynomial r of degree $m-1$. Hence V_m also reduces F, and $F|_{V_m}$ is the projection onto the normalized difference $(e_m - f_m)/\sqrt{2}$:

$$F|_{V_m} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{3}$$

Combining (2) and (3),

$$[P,Q]|_{V_m} = [E,F]|_{V_m} = \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}, \qquad \|[P,Q]|_{V_m}\| = \frac{1}{2}. \tag{4}$$

The vectors e_m are orthonormal as m varies because they lie in distinct homogeneous degrees, while (4) gives $\|[P,Q]e_m\| = 1/2$ for every m. An orthonormal sequence is weakly null, and a compact operator sends weakly null bounded sequences to norm-null sequences. Therefore $[P,Q]$ is not compact. $\square$

5 Verification, scope, and novelty

Verification. The argument uses only three exact facts: orthogonality of distinct monomials, equality of the two monomial norms under coordinate exchange, and the graded description of a principal submodule generated by a linear homogeneous polynomial. These hold in all three named modules. The two matrices in (2)–(3) multiply to the skew matrix in (4); no numerical or asymptotic estimate is involved.

Scope. The counterexample disproves the unrestricted "always" suggestion for every $n \geq 3$. It does not decide dimensions $n = 1, 2$. The zero varieties of z_1 and $z_1 + z_2$ meet on $\partial\mathbb{B}^n$ when $n \geq 3$, so the example does not contradict the source's positive disjoint-boundary-variety result.

Bounded novelty search. On 27 August 2026, searches used arXiv:1204.0620, the exact paper title and authors, the phrases "projections onto [p] and [q] almost commute" and "principal polynomial submodules projection commutator compact", and the specific pair $z_1, z_1 + z_2$. The searches found the source and later work on essential normality of individual polynomial-generated submodules, but no paper stating this counterexample or resolving this pairwise-projection question. This supports, but does not prove, novelty.

Human-review recommendation. Verify the restrictions of E and F to V_m, especially the orthogonality of $u_m - v_m$ to $(z_1 + z_2)\mathcal{H}_{m-1}$. Once that two-line calculation is confirmed, noncompactness is immediate.

References

[1] R. G. Douglas and K. Wang, *Some Remarks On Essentially Normal Submodules*, arXiv:1204.0620 (2012); published in *Concrete Operators, Spectral Theory, Operators in Harmonic Analysis and Approximation*, Operator Theory: Advances and Applications 236, Birkhäuser, 2014, pp. 159–170.